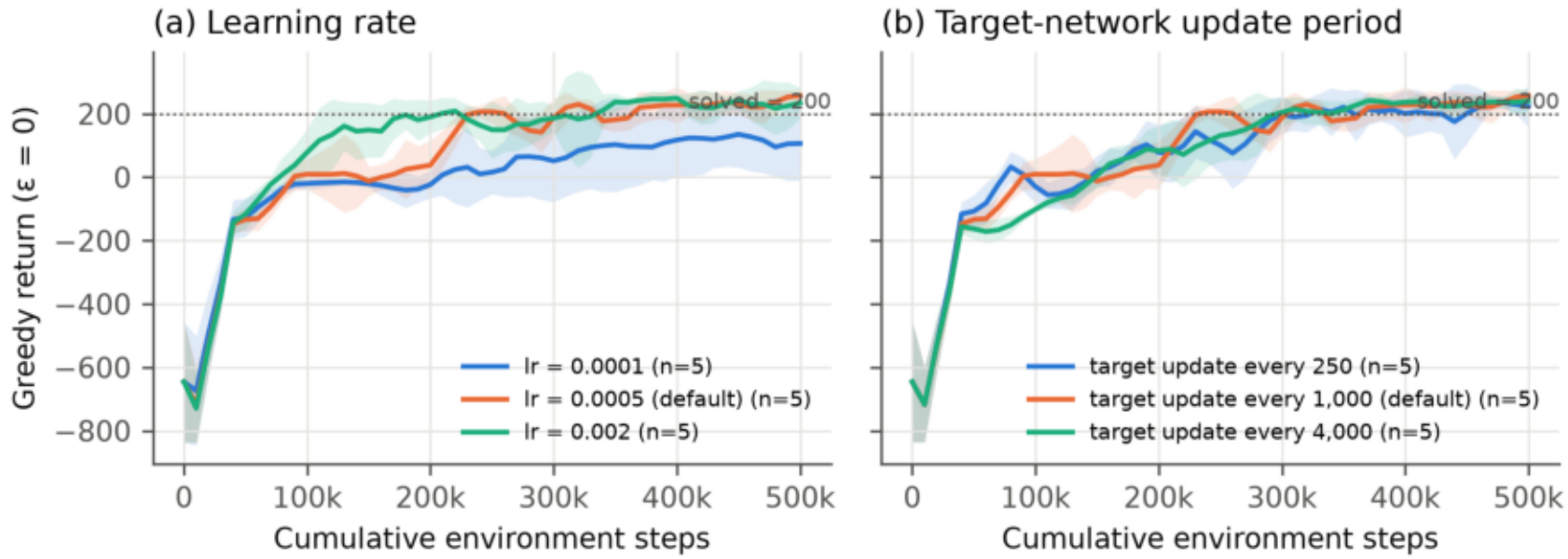
Tier 2 — Hyper-parameter study (LunarLander-v3)

One parameter at a time around the default · 500k steps · 5 seeds · greedy eval, 95% CI



Parameter	Value	Final	Drop	Steps to 200	Welch p
lr	0.0001	106.4 ± 116.6	83.5 ± 85.5	293k (3/5)	0.022 *
lr	0.0005 (default)	257.6 ± 17.4	14.7 ± 12.4	200k (5/5)	—
lr	0.002	234.4 ± 48.2	46.8 ± 49.6	118k (5/5)	0.26
target update	250	223.5 ± 64.4	53.3 ± 62.4	146k (5/5)	0.22
target update	1000 (default)	257.6 ± 17.4	14.7 ± 12.4	200k (5/5)	—
target update	4000	241.0 ± 33.4	33.4 ± 31.9	252k (5/5)	0.27

Takeaways

- Only lr = 1e-4 is significantly worse (p = 0.022): too slow for 500k steps.
- Higher lr / more frequent target copy → faster, but less stable (larger drop).
- Target every 4000 → slower but steady.
- Same speed-vs-stability trade-off as the ablation (no target = period of 1).

Official check — 100 greedy episodes on the FINAL weights

Thresholds are defined over 100 episodes · held-out start states · experiments/eval100.py

Env	Config	100-ep mean	Seeds ≥ thr.	Thr.
CartPole-v1	dqn_eps5k	300.2 ± 153.5	5/10	475
CartPole-v1	tabular_bins10	384.6 ± 87.5	4/10	475
CartPole-v1	tabular_bins6	393.7 ± 70.8	1/10	475
MountainCar-v0	dqn	-188.4 ± 17.9	0/10	-110
MountainCar-v0	dqn_long_eps	-174.5 ± 29.7	1/10	-110
MountainCar-v0	tabular_optimistic	-174.2 ± 11.4	0/10	-110
LunarLander-v3	double	233.8 ± 34.7	8/10	200
LunarLander-v3	dueling	212.0 ± 43.4	7/10	200
LunarLander-v3	full	226.9 ± 34.3	7/10	200
LunarLander-v3	no_replay	-442.2 ± 303.1	0/10	200
LunarLander-v3	no_target	188.2 ± 69.6	6/10	200

How to read it

- LunarLander full DQN passes: mean 227 ≥ 200, 7/10 seeds.
- CartPole: every seed reached 475 during training, but only 4–5/10 final policies keep it → catastrophic forgetting, measured on final weights.
- MountainCar: optimistic table finds the flag in 64% of episodes (DQN 19%), but in ~160 steps > 110 → it finds the goal, it does not 'solve' it.
- Negative results, explained.